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Roll No

ME-701(A) (GS)

B.E. VII Semester Examination, June 2020

Grading System (GS)

Design of Heat Exchanger

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Discuss classification of heat exchangers based on construction features.
2. Are we really getting extra advantage by providing Baffles in Shell and tube heat exchanger justify your answer.
3. a) Differentiate between parallel and counter flow heat exchangers.
b) What is overall heat transfer coefficient?
4. Water with a flow rate of 0.05kg/s enters an automobile radiator at 400K and leaves at 330K. The water is cooled by air in cross flow which enters at 0.75kg/s and leaves at 300K. If the overall heat transfer coefficient is 200W/m² K, what is the required heat transfer surface area?
5. Write short notes on :
 - a) Heat pipe
 - b) Software in heat exchanger design
 - c) Cooling and dehumidifying coils
 - d) Design of air washer
6. Write short notes (any two)
 - a) Hydraulic diameter and its impact on heat exchanger
 - b) Regenerator Vs Recuperator
 - c) Turbulence and friction factor in pipe flow
 - d) Effect of channel divergence
7. How are the TEMA charts useful in design of heat exchangers?
OR
What is Airwasher? Discuss it's design requirement.
8. Write short notes any two
 - a) Cross flow heat exchanger and its analysis
 - b) NTU method
 - c) Fouling factor
 - d) Testing of Heat Exchanger